

PARTS LIST

Semiconductors:

- IC1 - ATmega8A MCU
GPS1 - NEO 6M GPS module with antenna
LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-kilo-ohm
R2-R6 - 4.7-kilo-ohm
R7 - 1.5-kilo-ohm
R8 - 1-kilo-ohm
VR1 - 5-kilo-ohm potmeter

Capacitors:

- C1 - 1 μ F, 16V electrolytic
C2-C4 - 0.1 μ F ceramic disk

Miscellaneous:

- LCD1 - 16x1 LCD
CON1 - 2-pin terminal connector
CON2 - 4-pin connector
CON3 - 6-pin connector
S1-S3 - Tactile/push-to-on switch
SJ1-SJ2 - Shorting jumper
J1-J2 - Jumper wire
- 5V DC, 250mA power supply

location along with LCD1. Jumpers J1 and J2 shown in the PCB can be any conductor wires. Connect these before switching on the circuit.

Jumper setting. Connect SJ1 in 12-hour format, otherwise it will be set in 24-hour format by default. Connect SJ2 for UTC time, otherwise it will be set in local time (UTC + 5.30).

For example, the time stamp received by the GPS module is 14:15:16, which is in UTC time in 24-hour format by default. Now, LCD1 will display the time as per the settings for SJ1 and SJ2, as shown in the table, assuming the local time constant as + 05:30:00.

To change the local time, press minutes and hours switches (S2 and S3, respectively) during initialisation of the system when NEO 6M GPS message appears on LCD1. Change local time constant only once. **EFY**



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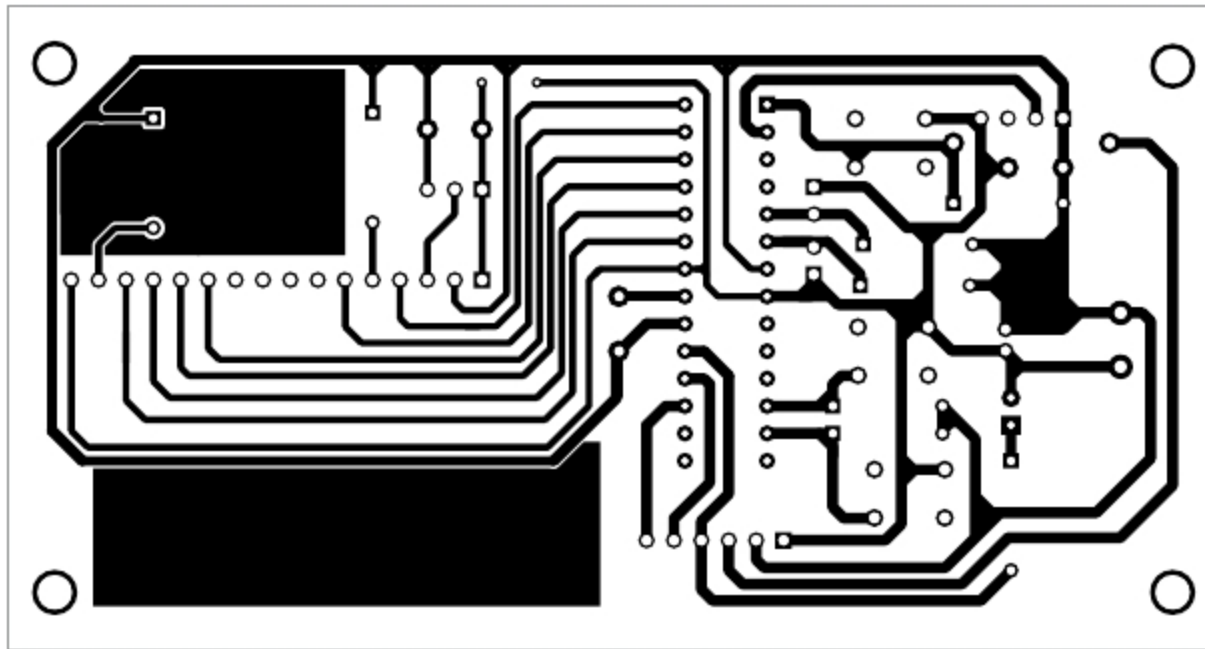


Fig. 2: Actual-size PCB layout of GPS clock

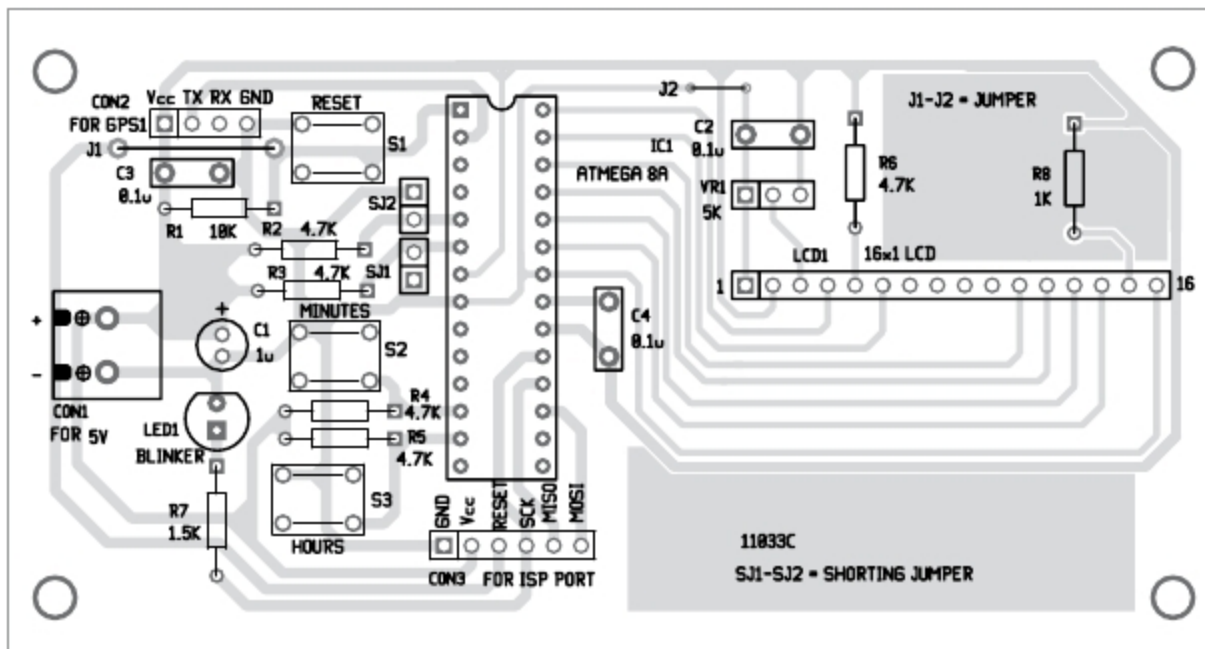


Fig. 3: Components layout for the PCB

JUMPER SETTINGS

SJ1 status	SJ2 status	Time display	Remarks
Disconnected	Connected	14:15:16	UTC in 24-hour format
Disconnected	Disconnected	19:45:16	Local time in 24-hour format; UTC + 05:30
Connected	Connected	02:15:16	UTC in 12-hour format
Connected	Disconnected	07:45:16	Local time in 12-hour format; UTC + 05:30

reception. This may take some time, say, three to five minutes. LCD1 will display "Connecting.." till the data is received from the satellite. Once the data is received, the time will be displayed on LCD1 in HH:MM:SS format continuously. In case GPS module is not connected, or is giving a connection error, 'NO GPS MODULE' message will be displayed on LCD1.

Use shorting jumpers SJ1 for 12/24-hour format and SJ2 for local/UTC time to be displayed. LED1 will indicate that the MCU is processing

the received data. Potmeter VR1 is used for adjusting the contrast for LCD1.

Construction and testing

An actual-size PCB layout of GPS clock circuit is shown in Fig. 2 and its components layout is shown in Fig. 3.

Assemble the circuit on the PCB and connect 5V across CON1 to operate the circuit. Enclose the circuit in a suitable box and connect GPS1 across CON2. Connect the antenna to GPS1. Place the unit at a suitable